FMI	ADMINISTRATIVE INFORMATION	FMI No: 25379

FMI Title: VCT Tube Pump Capacitor Replacement

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date: 1/13/06
☑ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date: 7/14/06

☑ **Staged FMI**

Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.

☑ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.

☐ **Parts Ordering**: GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator

☐ **No Parts** are required to complete this FMI.

☐ **Batch with** ______________________________

To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).

☐ **Non-Staged**

No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

Unit Tracked:

FMI Instructions Part No.: 5167225-100

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25379 **FMI Title:** VCT Tube Pump Capacitor Replacement

Customer Full Name:

Number and Street Address:

City and Country:

Country Code **Site Number** **System I.D.**

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1.) Modification Complete or previously modified.
2.) FMI not required per FMI instruction.
3.) Not modified. Unit in storage. FMI kit returned to stock.
4.) Modification refused or equipment altered. Customer signature and title required.
5.) Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer Comments

Effectivity

The following CT Equipment is affected: All VCT systems manufactured before February 2006 with the following gantry model numbers.

64 slice Gantry model # 5124069 32 slice Gantry model # 5126093

Related/Prerequisite FMI's

None

Purpose

The purpose of this FMI is to replace the current VCT Tube Oil Pump capacitor with a higher voltage rated capacitor. The current capacitor can be damaged due to higher voltage levels reached on VCT systems due to pump motor feedback. Failure of the current capacitor can cause Xray Tube failure in addition to the pump failure.

Furnished Materials

FMI 25379 Kit (P/N 5167069) contains the following materials:

Qty	Description	Part #
1	FMI Document	5167225-100
1	Pump support Bracket Assm. (Bracket with U-bolt Assm 5161681 and Harness/capacitor 5165908)	5165785

Tools Required

Qty	Description	Part #
2	Cable ties (ty-wraps)	Field Supplied
1	Wire Cutter	Field Supplied
1	Field Torque wrench kit	Field Supplied
1	5mm Allen wrench	Field Supplied
1	10mm and 11mm sockets with extension (3/8" drive)	Field Supplied
1	DVM	Field Supplied
AR	Loctite 242	Shipped with system

People Power

"1.5" hour labor and "1" hour of travel

Procedure

Preparation

1. Remove gantry side covers.
2. Disable Axial drive and HVDC from the Service Switch Panel.
3. Rotate the gantry so that the pump location is on the left (TGPU side) as shown in Figure 1 and disable gantry power using the Service Switch Panel.

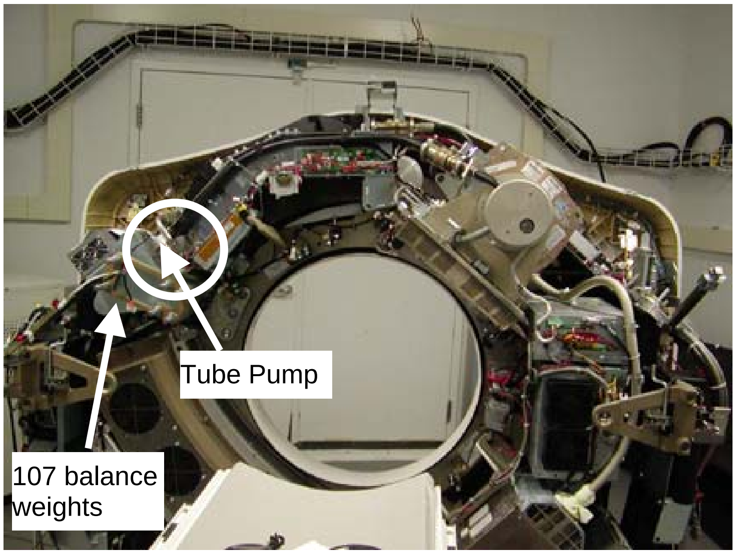

Figure 1 Gantry initial positioning

4. Verify the tube pump capacitor is mounted exactly as shown in Figure 2. If there is a capacitor mounted in a different location, then do NOT perform this FMI, the system has already been modified. Close this FMI for this system as previously modified.

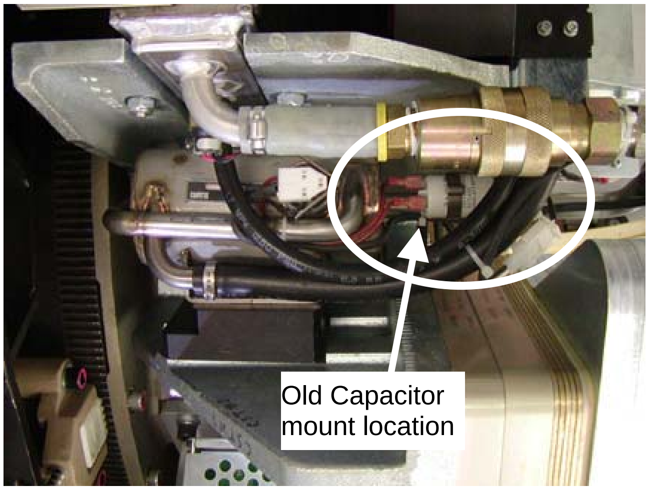

Figure 2 Old Capacitor Mounting Location

5. Use LOTO procedures to make sure all power is off to the gantry.
6. Remove gantry top and front covers.

Removing Old Harness

7. Locate the pump electrical harness. Disconnect the pump harness connector from the heat exchanger harness connector J52 (see Figure 3) on the front side of the gantry. This connector may be found ty-wrapped on the pump tube or front side of the heat exchanger.

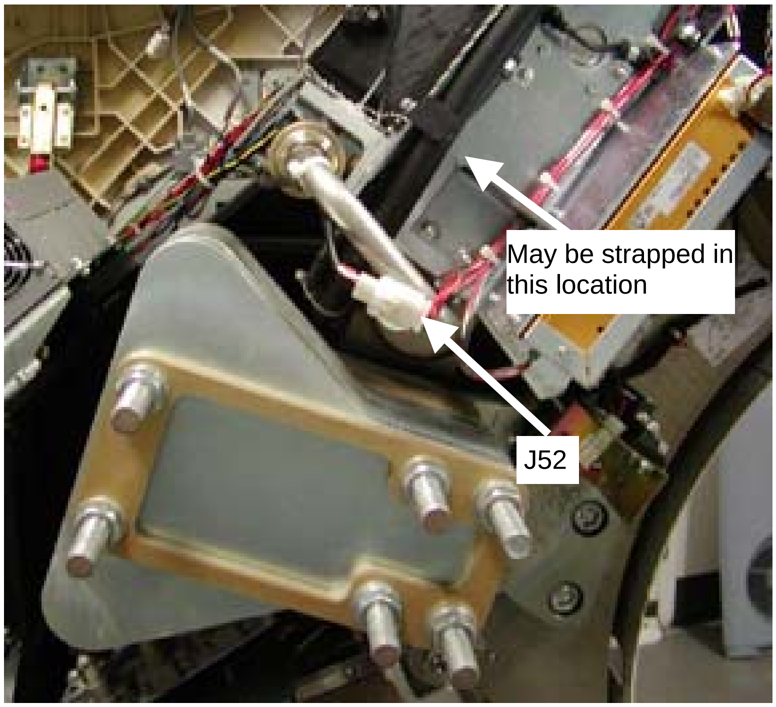

Figure 3 Pump connector J52

8. Disconnect the pump harness connector from the pump connector pins. Connector lifts up away from pump.

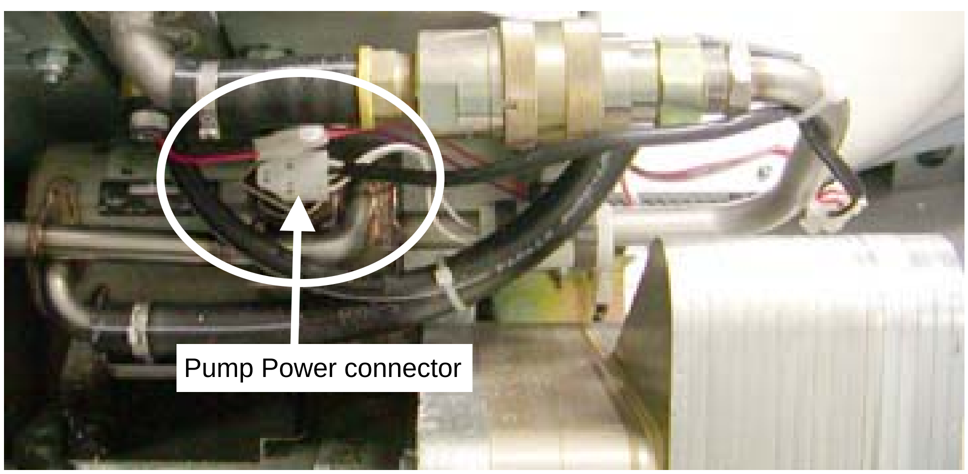

Figure 4 Pump Power connection

9. Check the pump resistances by measuring the resistance between pump connector pins per Table 1. Replace the pump if any of the resistances fail to meet the specification. (Refer to Figure 5 for pump connector pin locations)

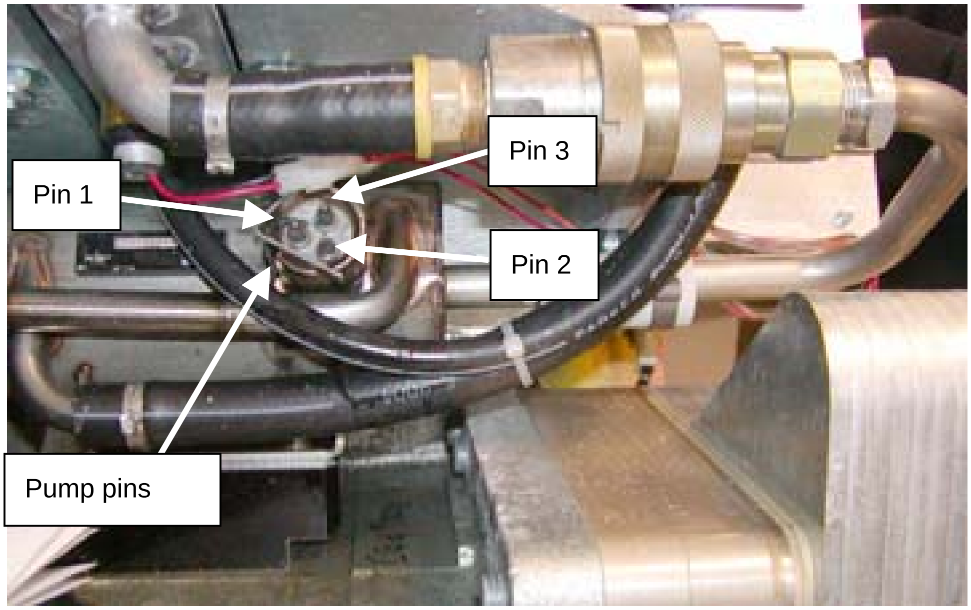

Figure 5 Pump Power Connector Pins

Table 1 Pump Resistance

Pins	Specification
Pin 1-3	24 – 30 ohms
Pin 2-3	12.5 – 18 ohms
Pin 1-2	39 – 46.5 ohms
Pin 1,2,3 to ground (pump casting is gnd)	Open (Infinite Resistance)

NOTE: If any resistance check fails, prevent further X-Ray generation with the system.

DO NOT RUN SCANS, SYSTEM IS DOWN UNTIL PUMP IS REPLACED.

Order and replace the tube pump per Table 2 but continue with steps 10-12 now to check the Tube Stator in case damage has already occurred. **DO NOT CONTINUE PAST STEP 12**.

Table 2 Tube Pump Part Numbers

If pump 5105346-2 is available (**Pump with new Capacitor**)	Order/Install new pump. After installation close this FMI as already done since new pump will have new capacitor.
If only pump 5105346 is available (Pump with old capacitor)	Install replacement old style pump and perform this FMI from the start.

10. Disconnect the tube stator connector J1 from the Aux Box.

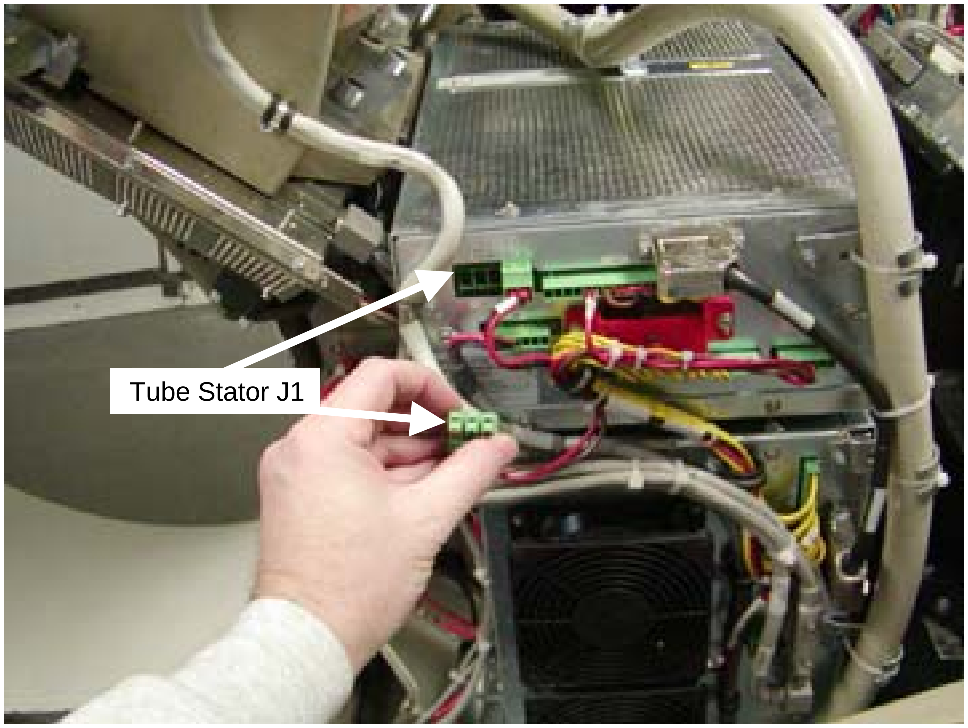

Figure 6 Tube Stator Harness

11. Measure the resistance between the stator connector pins per Table 3. Check Pin to Pin and Pin to Ground. Resistance lower than specification is an indication of a shorted winding.

Table 3 Tube Stator Resistance

Pins	Specification
Pins 1-2	2.0 – 3.0 ohms
Pins 2-3	2.0 – 3.0 ohms
Pins 1-3	2.0 – 3.0 ohms
Pin 1,2,3 to Ground	Open (Infinite Resistance)

NOTE: If the resistance check fails, the tube must be replaced. Complete this FMI and order a Tube, D3194T. Permit customer to use system until tube arrives if there have been no recent aborted scans due to rotation errors.

12. Reconnect the Aux Box J1 connector.

13. Back at the Tube Pump, cut the cable ties holding the existing Tube pump harness wires in place.

14. To remove the existing capacitor harness, cut the capacitor leads approx. 6mm – 18 mm (¼" to ¾") from the capacitor end and remove the old harness from the gantry.

IMPORTANT: Do **NOT** try to remove the old capacitor. This can result in cracking the tube weld on the pump.

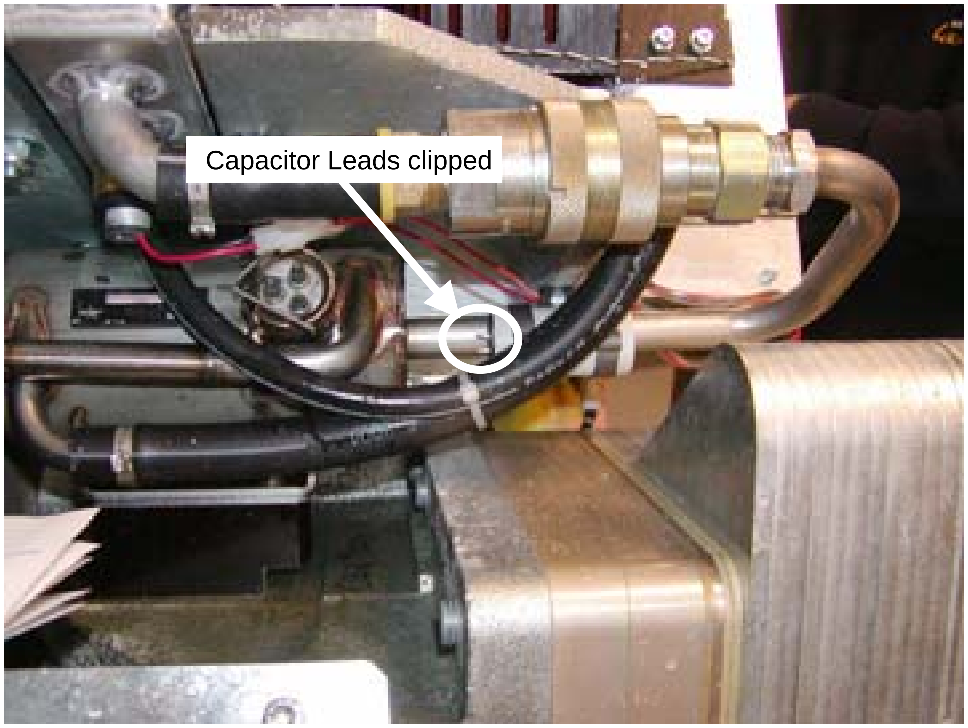

Figure 7 Capacitor Leads clipped

Installing New bracket with capacitor and Harness

15. At the heat exchanger, using a 10mm socket and extension, remove the M6 nuts and washers (leave the screws in place) from the two socket cap screws at the heat exchanger location shown in Figure 8. The nuts/washers will be used in the next step. You may need a 5mm Allen wrench on the backside to keep the bolts from spinning.

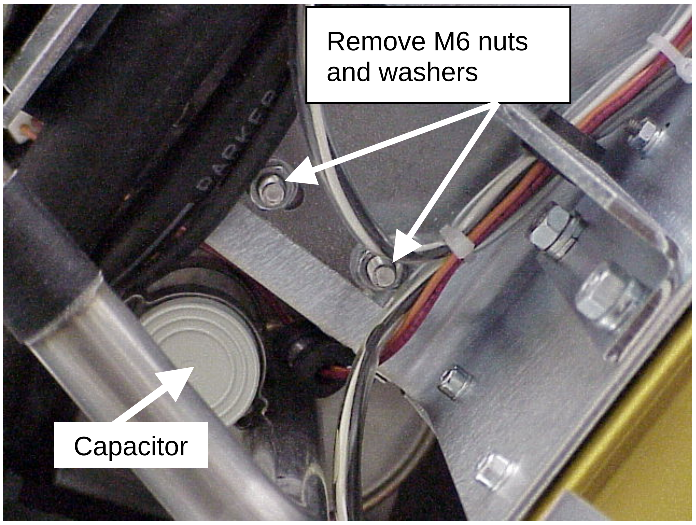

Figure 8 Bracket mount location

16. Mount the Bracket assembly as follows, use Figure 9 for reference.
    a. Remove U-bolt from the new bracket assembly.
    b. Mount the bracket on the Heat exchanger using the M6 nuts and washers removed in step 15 and hand tighten the nuts. Make sure no wires are pinched between bracket and Heat exchanger or tubing.
    c. Place the U-bolt around the pump tube and thru the bracket slots as shown in Figure 9. Install the washer plate and hand tighten the nuts to hold the bracket in place.

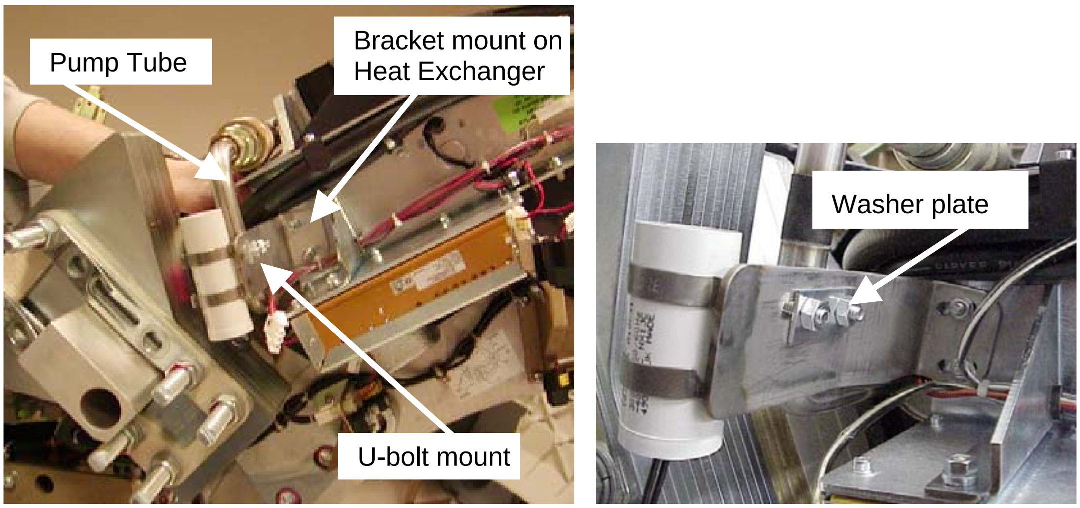

Figure 9 Capacitor Mount

17. Tighten the two nuts at the heat exchanger. Use a 5mm Allen Wrench on the Bolt heads if necessary. Torque to the value in the Table below.

9.9 N-m	87.6 lb-in	7.3 lb-ft	101 Kg-cm

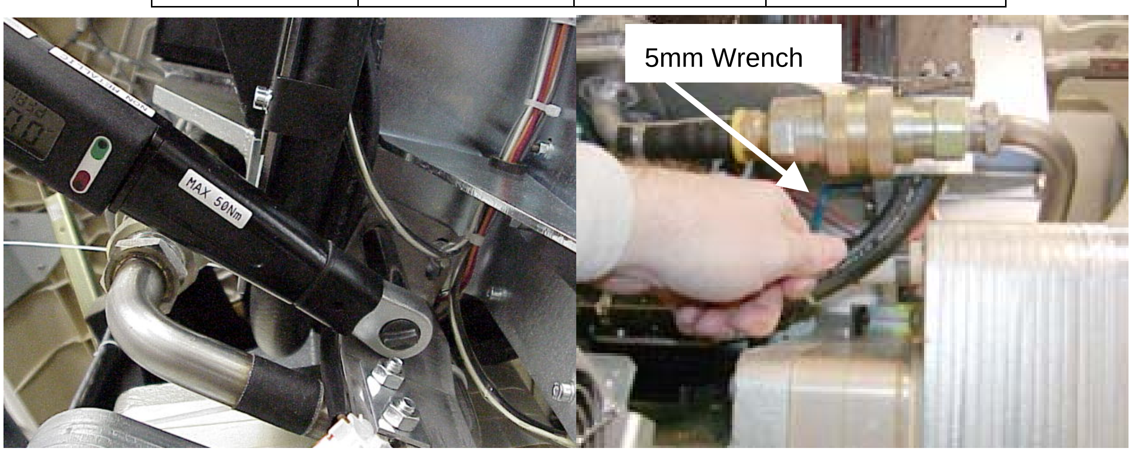

Figure 10 Torque Bracket mount

18. Remove U-bolt nuts one at a time, apply Loctite 242. Tighten the nuts evenly such that there is about the same length of threads sticking out of the nuts. This ensures that the U-bolt is not twisted or tilted and there is even pressure on the tube pump pipe.

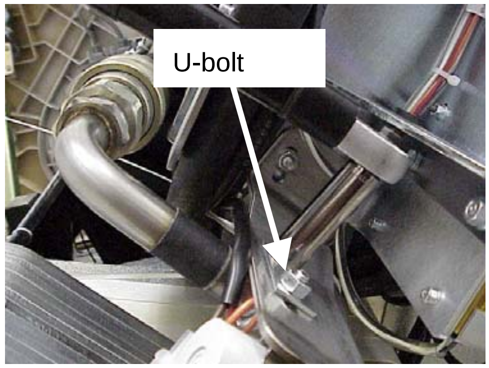

Figure 11 U-bolt mount nuts

19. Torque the two nuts at the U-bolt to the value in the Table below.

8.0 N-m	70.8 lb-in	5.9 lb-ft	81.5 Kg-cm

20. Route the harness under the black pump hoses as shown in Figure 12. Connect the pump wire harness connector to the Heat Exchanger (J52) and to the pump power connector. Use cable ties to strap the wiring to hoses/tube in the two locations shown in Figure 12. Cable ties should be snug but not pinching wires or tubing and wires to connectors should **not** be pulled tight. The tubes will expand and contract due to oil temperature. Cut off any cable tie excess.

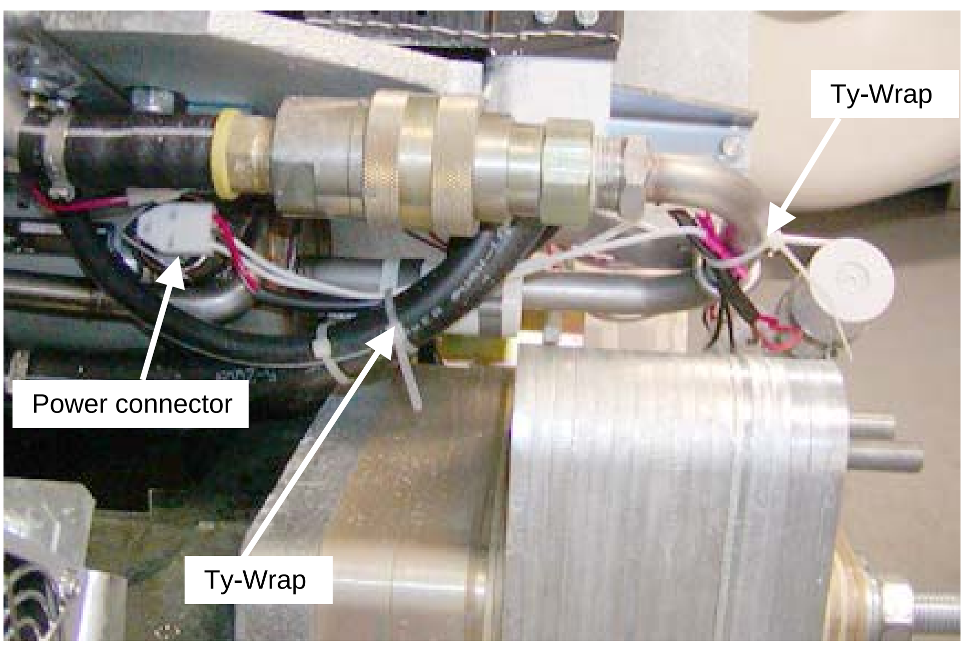

Figure 12 Harness Routing

21. Install any other ty-wraps taken off when removing the old harness. Make sure the new harness is secured and will not shift during rotation.
22. Remove LOTO equipment and enable the Gantry 120VAC switch from the Service Switch Panel. Verify that the tube pump is functioning correctly. Pump motor and oil flow should be heard as a hissing sound or you can pinch one of the tube flexible hoses with your fingers to feel the oil flow.
23. Rotate the gantry by hand and visually inspect to verify no loose components.
24. Disable Gantry power using the Service Switch Panel and reinstall the gantry covers (Top cover fans should not be connected with power on).
25. Enable Gantry Power, HVDC and Axial Drive from the Service Switch Panel and enable table drive.
26. Install right side cover.

Finalization

1. From the **Common Service Desktop**, **Calibration menu** select **[Gantry Service balance]** and perform a balance check. Rebalance the gantry if necessary. Should only require balancing if gantry was almost out of specification to start.

2. Perform a test scan using the 20cm QA phantom to verify system operation.

3. Close the FMI.